PLamb Cosmology Project — Update Summary

1. Code Updates (10V6_plus)

Recent updates to the 10V6_plus.py program introduced several new modelling capabilities:

• Fluctuating Λ(z): stochastic or time-varying cosmological constant with support for white noise, Ornstein–Uhlenbeck (OU), piecewise, and spline realizations.

• Λ–A_acc Coupling: allows explicit interaction between accretion-driven acceleration terms (A_acc) and stochastic Λ fluctuations.

• Discrete-Event Integrator: integrates cosmological distances in discrete event-steps (Δz) rather than continuous Simpson's rule.

• Expanded Variable-c(z) Modelling: functional families include linear_z, log1pz, piecewise, density-coupled, power1pz, expz, saturating, poly_log1pz, and cheb_log1pz.

• Extended BAO/Planck handling: now allows c(z) to propagate into DH(z), R, and r_s.

• PBH (Primordial Black Hole) Diagnostics: entropy, holographic projection, Fourier spectral modes, HHT analysis.

• Noether Symmetry Testing: hard freeze or Gaussian soft penalties on key parameters (A_acc, n_acc, γ_c, ε_M).

2. Key Equations

Baseline distance measures (generalized with $\Lambda(z)$):

- $E(z) = \mathrm{sqrt}(\Omega m (1+z)^3 + \Omega\Lambda(z) + \Omega k (1+z)^2)$
- $D\_C(z) = \int dz' / E(z')$ (with Simpson or discrete-events)
- $D\_L(z) = (1+z) D\_C(z)$

Stochastic $\Lambda(z)$:

- $\Omega\Lambda(z) = \Omega\Lambda + \delta\Lambda(z)$
  - OU: $\delta\Lambda(z+\Delta z) = \delta\Lambda(z) \exp(-\Delta z/\Delta z\_corr) + \eta\ \sigma\sqrt{(1-\exp(-2\Delta z/\Delta z\_corr))}$
  - White noise, spline, or piecewise controls supported.

Coupling Λ–A_acc:

- $\delta\Lambda(z) \rightarrow \delta\Lambda(z) + \gamma \cdot A\_acc(z)$

PBH entropy proxy:

- $S \sim A/4\ell\_p^2 \propto (GM/c^2)^2$ (linked to photon wavelength at horizon diameter).

3. Acronyms and Variables

H0: Hubble constant today
Ωm: Matter density fraction
ΩΛ: Cosmological constant density fraction
Ωk: Curvature density fraction (1−Ωm−ΩΛ)
A_acc: Accretion-driven acceleration amplitude
n_acc: Accretion acceleration index (power-law scaling)
γ_c (gamma_c): slope parameter for c(z)
ε_M (epsilon_M): mass-coupling parameter
Λ(z): stochastic cosmological constant
Δz: event-size step for discrete integration
r_d, r_s: sound horizon at drag/decoupling epoch
l_A, R: Planck distance priors
PBH: Primordial Black Hole
OU: Ornstein–Uhlenbeck process
HHT: Hilbert–Huang Transform

4. Investigations to Date

• Baseline FR (Flat Relativistic) model runs against Pantheon+SH0ES SN data, with and without BAO/Planck priors.

• Kerr–de Sitter (KdS) tests: acceleration attributed to centripetal artifacts of rotating space-time fabric.

• ΛCDM comparisons: standard cosmology baseline with Ωm, $\Omega\Lambda$ fitted.

• PBH/holographic tests: exploring entropy growth from accretion and its imprint on galactic distribution.

• Noether symmetry probes: enforcing conservation (hard) or testing soft-breaking penalties.

• Variable-c fits: including piecewise and density-coupled families.

5. Notable Results

• FR non-expanding baseline fits show consistency with SN data but differ in BAO handling.

• Kerr–de Sitter framework yields best-fit values around $H0 \approx 71.2$, $\Omega m \approx 0.279$, $\Omega\Lambda \approx 0.739$, $A\_acc \approx 0.030$.

• PBH entropy scaling indicates potential links between galaxy alignment and horizon-scale holography.

• Noether soft penalties allow testing of symmetry-constrained cosmologies with moderate evidence for departures.

• Fluctuating $\Lambda(z)$ introduced as a new avenue; pending runs will test amplitude and correlation length sensitivity.

6. Planned / Pending Runs

• OU $\Lambda(z)$ sweeps across amplitude (10^{-122} to 10^{-119}) and correlation length ($\Delta z = 0.05$–0.6).

• Discrete-event integrator comparisons against Simpson to look for residual structure.

• Λ–A_acc coupling scans to test if accretion-driven acceleration correlates with fluctuating dark energy.

• BAO $c(z)$ vs $c0$ comparisons to test light-speed variability signatures.

• Planck distance prior variants with $c(z^*)$ included.

• PBH standalone diagnostics to identify possible holographic or spectral node structures.

• Novelty scans of SN residuals (periodicity in $\log(1+z)$, binned RMS, runs tests).